Lesson 9.4 Guided Notes

Session 1 · Substitute and Solve

I can substitute a value into an equation to find the matching output. · 6.EE.C.9

Word / Spanish	Meaning — complete the blank	Example
substitute / sustituir	Put a known number in _____ of a letter.	put 8 in for x
order of operations / orden de las operaciones	_____ first, then add or subtract.	3 × 4 + 10 = 22
reasonable / razonable	Makes _____ for the story.	17 days, not 17 years

Key idea

To find an output, _____ the input for the variable.

With a fee, _____ first, then add the fee.

Worked example

A bakery makes 24 rolls each hour, so r = 24h. Is 5 hours enough for an order of 150 rolls?

Substitute h = 5: r = 24 × _____.

Multiply: r = _____ rolls.

Compare: 120 _____ 150.

Answer: _____, 5 hours makes only 120 rolls.

You try

1. Substitute to find y.
a. y = 4x + 7; x = 6 _____
b. y = 12x; x = 9 _____

_____

2. A rental costs c = 8h + 15 dollars for h hours. Find the cost for 5 hours.

_____

Sum it up: When I substitute into an equation with a fee, I ___ before I ___.

_____

Session 2 · Working Backward to a Goal

I can work backward from a goal to find the input an equation needs. · 6.EE.C.9

Word / Spanish	Meaning — complete the blank	Example
work backward / trabajar hacia atrás	Start with the _____ and undo the steps.	goal first, then divide
inverse operation / operación inversa	An action that _____ another action.	÷ undoes ×
at least / al menos	That number _____.	at least 42 cars

Key idea

Put the goal in place of the _____, then solve for the input.

With a fee, _____ the fee first, then divide by the rate.

If only whole items make sense, round _____ to reach the goal.

Worked example

Tickets cost $6. A club needs at least $100. How many tickets must it sell?

Put in the goal: 100 = _____n.

Divide: 100 ÷ 6 ≈ _____.

16 tickets make $96 (too low); _____ tickets make $102.

Answer: The club must sell _____ tickets.

You try

1. Solve for x.
a. 11x = 528 _____
b. 4x + 10 = 50 _____

_____

2. A van holds 8 students. 60 students are going on a trip. How many vans are needed?

_____

Sum it up: When I work backward with a fee, I ___ first because ___.

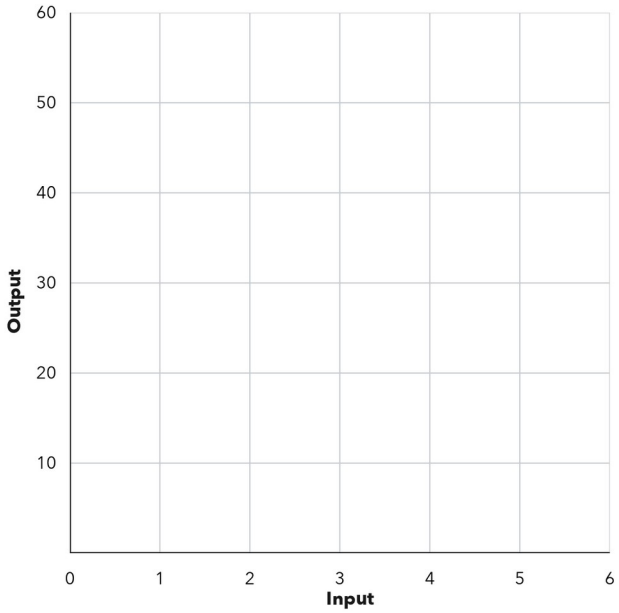
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Session 1 · Model, reason, and explain

Substitute and Solve · 6.EE.C.9

Success: represent the quantities, justify the relationship, and check the result with the correct units.

A rental costs c=7h+12. Is $40 enough for 4 hours? Show substitution and a contextual decision.



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: adds the fee before multiplying or gives a value without answering the decision.. Explain what changes and what stays true.

_____

_____

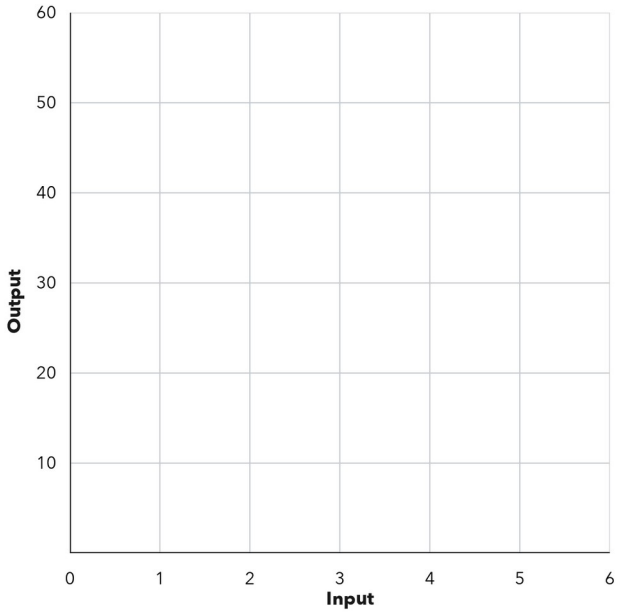
Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Working Backward to a Goal · 6.EE.C.9

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Tickets sell for $6. Find the fewest whole tickets needed to raise at least $100. Separately, find the most tickets that can be bought with at most $100. Test neighboring integers.



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: always rounds up, including a maximum-budget problem.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?